

AI Strategy and Transformation

Book 9 Lesson 1

Executive and Technical Learning Edition

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Mission-led strategy, evidence-based portfolio choices, responsible governance and accountable transformation.

Professional provenance

This lesson draws on Mbaye Kebe's international consulting experience in official statistics, agricultural censuses, household surveys, information-system design, results-based management, monitoring and evaluation, data warehousing, business intelligence and analytical modernization. Experience-derived cases reflect work with FAO, national statistical institutions, ministries and development programmes across Africa, Asia, Latin America, Europe and North America. They are teaching cases: historical facts are distinguished from contemporary AI applications.

Purpose and central conclusion

AI transformation succeeds when institutions treat it as governed organizational change. Models and platforms create durable value only when mandates, evidence, decision rights, people, financing and learning mechanisms are aligned. The executive task is to authorize a portfolio that the institution can operate responsibly, not to maximize the number of pilots.

Learning outcomes

- Formulate an AI strategy from mandate, beneficiaries and measurable outcomes.
- Assess readiness with evidence and uncertainty.
- Prioritize initiatives using value, feasibility, risk and dependencies.
- Design governance, decision rights and stage gates.
- Build a phased roadmap and benefits framework.
- Prepare a defensible executive strategy recommendation.

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How to use this lesson

Executives may read Sections 1, 4, 7, 9, 10, 12 and 14 as a decision pathway. Technical and programme teams should complete the full lesson and execute the companion notebook. Each section ends with an executive implication that can be transferred into an institutional strategy record.

1 AI strategy is institutional strategy

1.1 Begin with mandate and value

An AI strategy is a set of institutional choices about where intelligence should improve decisions, services and accountability. It is not a catalogue of tools. The starting point is the mandate: what the institution must deliver, for whom, under which legal and operational constraints. A credible strategy therefore names beneficiaries, establishes a baseline, defines measurable outcomes and identifies the executive who owns each result. Technology enters only after the decision problem is clear.

This orientation is especially important in public institutions and official statistics. A technically impressive model can still be strategically irrelevant if it does not improve a mandated output, reduce a material risk, expand equitable access or strengthen an accountable decision. Mission alignment is a necessary condition, not a decorative score added after a use case has already been selected.

1.2 Distinguish transformation from acquisition

Procurement asks what to buy. Transformation asks what must change in processes, roles, evidence, controls and capabilities. Buying a platform may be part of the answer, but it cannot substitute for redesigned decision rights, qualified data, trained counterparts or operational ownership. The strategy must describe a future operating state and the sequence by which the institution can reach it.

Executives should demand a transformation thesis in plain language: the institutional constraint, the proposed mechanism, the expected benefit, the evidence required, the risks introduced and the conditions under which the initiative will stop. This creates a disciplined bridge from ambition to authorization.

1.3 The executive strategy test

A useful strategy survives six questions. Which outcome changes? What evidence establishes the baseline? Why is AI necessary? Who owns the benefit and the risk? What capability must exist before scaling? What would cause leadership to stop or redesign the initiative? If these questions cannot be answered, the institution has an idea rather than a strategy.

Scores and dashboards structure deliberation, but they do not confer authority. The responsible executive retains the duty to interpret uncertainty, balance competing objectives and record the decision. SRAI treats this recorded reasoning as part of the production evidence.

Executive implication

Record the decision, evidence, owner, uncertainty and next review date. Strategy becomes operational only when these elements are explicit.

2 Lessons from institutional modernization

2.1 The continuity from information systems to AI

Three decades of work with censuses, surveys, monitoring systems and data warehouses show that AI transformation inherits earlier institutional realities. Definitions must be harmonized. Data must be timely and comparable. Regional and central offices require workable responsibilities. Users need trusted access. Managers need reports at operational, tactical and strategic levels. AI adds new analytical possibilities, but it does not repeal these requirements.

The practical lesson is that an institution becomes AI-ready by strengthening the whole evidence chain. Collection, validation, processing, storage, dissemination, interpretation and feedback must work together. A weakness at any stage becomes model risk downstream.

2.2 National ownership is a design requirement

Systems last when counterpart teams can operate, question and improve them. Training is therefore not an activity placed at the end of a project; it is a design constraint from the beginning. Documentation, reproducible analysis, transparent assumptions and maintainable architectures are strategic assets. A solution that cannot be governed locally creates dependency rather than transformation.

Consulting experience in Africa, Asia and Latin America repeatedly demonstrates that institutional context matters as much as software. Staffing rules, procurement cycles, connectivity, languages, budget horizons and inter-agency authority determine what can be sustained. The roadmap must reflect these constraints explicitly.

2.3 Diagnose before prescribing

A diagnostic mission should map mandates, stakeholders, decisions, data flows, systems, skills and risks before recommending technology. Interviews and document review are useful, but evidence must also include observed workflows, samples of data, system inventories, quality reports and actual decision products. The gap between formal procedure and operational practice is often where transformation succeeds or fails.

The baseline is not a judgement on staff. It is a shared representation of constraints and capabilities. When validated with counterparts, it becomes the reference against which options, investments and progress can be evaluated.

Executive implication

Record the decision, evidence, owner, uncertainty and next review date. Strategy becomes operational only when these elements are explicit.

3 Establishing the strategic baseline

3.1 Six evidence domains

The SRAI baseline covers mission and value, data and evidence, technology and delivery, responsible governance, people and adoption, and measurement and learning. Each domain is scored only when evidence exists. Missing evidence is not silently converted into an average value; it is recorded as uncertainty and may block authorization.

A maturity score is meaningful only when accompanied by its evidence register. The register identifies the source, owner, date, confidence and limitation of every material judgement. This makes the diagnostic reviewable and prevents false precision.

3.2 Map decisions and information flows

Process maps should identify where decisions occur, which information enters, who validates it, how exceptions are handled and what record remains. In a census programme, for example, field collection, quality control, correction, estimation and release have different owners and risk profiles. AI opportunities must be located within that chain rather than attached to it generically.

Decision mapping also exposes bottlenecks that do not require AI. A delayed approval, missing identifier or incompatible classification may be solved by governance, workflow redesign or standardization. Choosing the simplest adequate intervention protects resources and reduces risk.

3.3 Calibrate ambition to readiness

Strategic ambition can exceed present capability, but near-term commitments cannot. A roadmap should therefore separate foundational investments, controlled experiments, scalable services and longer-term options. High-value initiatives with weak data or governance remain in discovery until their dependencies are addressed.

This discipline avoids two common errors: refusing all experimentation until every foundation is perfect, and scaling attractive pilots before the institution can operate them safely. Controlled learning occupies the space between those extremes.

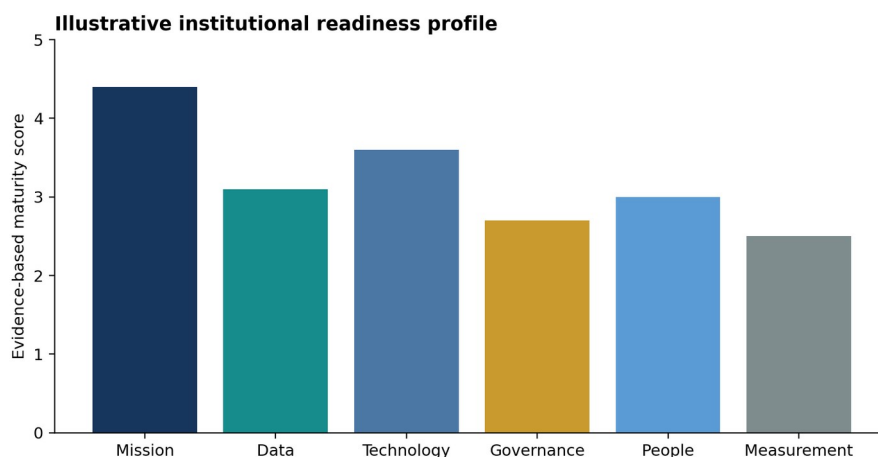


Figure 1. A readiness profile should expose bottlenecks rather than hide them in one average.

Executive implication

Record the decision, evidence, owner, uncertainty and next review date. Strategy becomes operational only when these elements are explicit.

4 The SRAI transformation framework

4.1 Mission and public value

The mission pillar connects each initiative to a declared outcome and beneficiary. It tests strategic relevance, distributional effects and opportunity cost. The institution must explain not only who benefits, but who might be excluded or burdened.

Evidence includes strategic plans, service standards, programme results frameworks and baseline indicators. Leadership should reject use cases whose benefits cannot be stated independently of the technology.

4.2 Data technology and delivery

The data pillar examines availability, quality, representativeness, lineage, access and stewardship. The technology pillar examines architecture, security, integration, monitoring and operational support. Delivery capability covers product management, domain expertise, testing and change control.

These pillars are interdependent. High-quality data without a maintainable delivery environment will not produce durable value; capable infrastructure without governed data can accelerate error.

4.3 Governance people and learning

Responsible governance defines authority, safeguards, review and recourse. People and adoption address leadership literacy, role design, training, incentives and national ownership. Measurement and learning connect baselines to outcomes, incidents, benefits and revision decisions.

Together, the six pillars create a balanced strategy. None can be treated as a residual category after the model has been built.

Pillar	Executive evidence
Mission and value	Mandate, beneficiaries, baseline, outcome owner
Data and evidence	Quality, lineage, stewardship, access
Technology and delivery	Architecture, security, integration, support
Responsible governance	Authority, safeguards, review, recourse
People and adoption	Skills, roles, incentives, ownership
Measurement and learning	Indicators, incidents, benefits, revision

Executive implication

Record the decision, evidence, owner, uncertainty and next review date. Strategy becomes operational only when these elements are explicit.

5 Diagnosing AI readiness

5.1 Readiness maturity levels

SRAI uses five descriptive levels: initial, emerging, defined, managed and adaptive. Levels describe observable capability rather than institutional prestige. An organization may be managed in data production yet emerging in AI risk governance. The profile matters more than a single average.

Readiness assessment should combine documents, system evidence, data samples, interviews and demonstrations. Confidence is scored separately so that a precise-looking result does not conceal weak evidence.

5.2 Weighting and sensitivity

Weights express strategic judgement. They should be approved, documented and tested. If a small change in weights reverses the priority order, the executive should see that instability rather than receive one definitive ranking. Sensitivity analysis is therefore part of governance.

The companion notebook recalculates rankings under alternative weight sets and simulates uncertainty. It identifies robust priorities, contested initiatives and evidence gaps requiring further investigation.

5.3 Interpret bottlenecks

A low governance score may block high-impact autonomous decisions while allowing low-risk knowledge retrieval. Weak interoperability may delay cross-agency applications but not a bounded internal pilot. Readiness decisions should be use-case-specific while remaining consistent with enterprise controls.

The output of readiness assessment is an action plan: which capability must improve, who owns it, what evidence demonstrates completion and which decision gate it unlocks.

Level	Observable condition
1 Initial	Activity is ad hoc and evidence is incomplete.
2 Emerging	Pilots and partial controls exist but vary by team.
3 Defined	Standards, roles and repeatable processes are documented.
4 Managed	Performance and risk are monitored and acted upon.
5 Adaptive	Evidence drives continuous improvement and retirement.

Executive implication

Record the decision, evidence, owner, uncertainty and next review date. Strategy becomes operational only when these elements are explicit.

6 Discovering AI opportunities

6.1 Start with a decision or service problem

An opportunity statement specifies the current decision, user, delay, error, cost or service gap. It records the baseline and explains the mechanism by which AI could improve performance. This prevents solutions from searching for problems.

For official statistics, appropriate opportunities may include coding assistance, anomaly detection, document retrieval or quality triage. Publication decisions, methodological approval and official estimates remain under accountable professional authority.

6.2 Test whether AI is necessary

Rule-based automation, database repair, process redesign or improved training may solve the problem more reliably. AI is justified when pattern recognition, language processing, prediction or complex prioritization creates material value that simpler methods cannot deliver at acceptable cost and risk.

The opportunity canvas requires a non-AI comparator. This makes incremental value visible and supports a defensible build, buy or do-not-proceed decision.

6.3 Define the value hypothesis

A value hypothesis states the expected change, affected population, time horizon and evidence method. It also names potential harms and distributional effects. A pilot validates this hypothesis; it is not merely a technical demonstration.

Benefits should have owners. Without an operational leader responsible for adoption and realization, projected savings or service improvements remain assumptions.

Executive implication

Record the decision, evidence, owner, uncertainty and next review date. Strategy becomes operational only when these elements are explicit.

7 Prioritizing the portfolio

7.1 Multi-criteria assessment

Initiatives are assessed against mission value, beneficiary reach, data readiness, technical feasibility, institutional capacity, risk acceptability, cost and time to evidence. Criteria definitions and scoring anchors are published with the result.

The total score is a navigation aid. Mandatory legal, ethical or security constraints remain gates, not tradeable points. A high value score cannot compensate for an unacceptable rights or safety risk.

7.2 Dependencies and portfolio balance

Foundational initiatives may score lower on visible short-term benefit yet enable several services. Dependency mapping prevents them from being crowded out by attractive front-end pilots. The portfolio should balance foundations, quick evidence, strategic bets and capability building.

Executives should also limit concurrent work. Too many pilots fragment scarce domain and governance capacity. A smaller portfolio with named owners and evidence dates is more credible than a long innovation list.

7.3 Portfolio decision records

Every selection should record the evidence considered, assumptions, dissent, conditions, accountable decision maker and review date. Rejected and deferred initiatives also need reasons. This protects institutional memory and improves later evaluation.

The portfolio is reviewed as evidence changes. Continuation is not automatic; initiatives may advance, pause, redesign or retire.

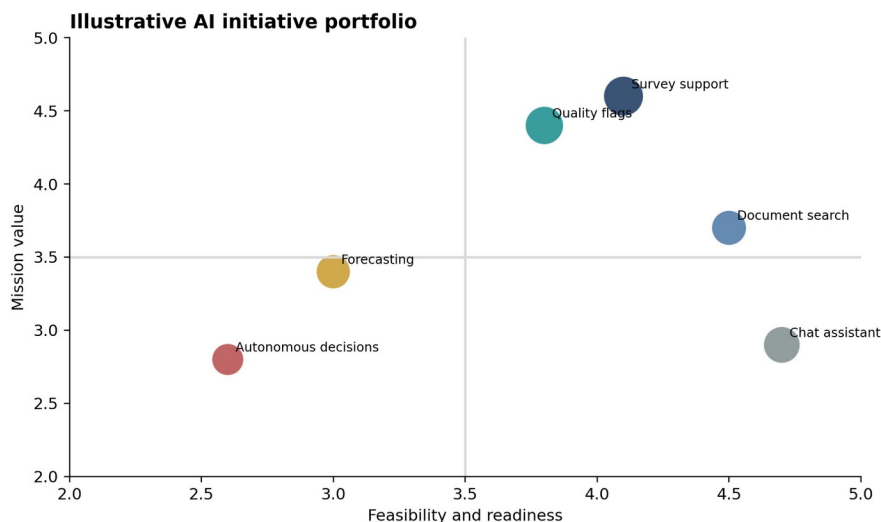


Figure 2. Portfolio position combines value with feasibility; risk gates remain separate.

8 Data and information foundations

8.1 Quality before scale

Completeness, validity, consistency, timeliness, uniqueness and representativeness affect different AI risks. Aggregate quality scores can hide critical subgroup failures. Assessment therefore follows the intended decision and population.

Survey and census experience reinforces the importance of classifications, metadata, frame quality, edit rules and reproducible transformations. Model development must preserve that lineage.

8.2 Architecture and interoperability

A warehouse, lakehouse or exchange platform creates value when ownership, standards and access rules are clear. Architecture should separate source systems, governed integration, analytical products and public dissemination. Interfaces must be monitored and versioned.

AI services should not create uncontrolled copies of sensitive data. Access, retention, encryption, logging and environment separation belong in the strategic design.

8.3 Stewardship and evidence lineage

Data owners authorize purpose and access; stewards maintain definitions and quality; technical custodians operate platforms; users remain responsible for appropriate interpretation. These roles must be explicit.

Lineage links a decision product back to source, transformation, model version, review and authorization. It enables audit, correction and learning.

Executive implication

Record the decision, evidence, owner, uncertainty and next review date. Strategy becomes operational only when these elements are explicit.

9 Governance risk and accountability

9.1 Decision rights

Governance identifies who proposes, validates, approves, operates, monitors and retires an AI system. Committees are useful only when their authority and escalation thresholds are clear. The accountable executive cannot delegate responsibility to a vendor or model.

Risk ownership should sit with leaders able to change the relevant process or accept residual exposure. Technical teams provide evidence; they do not make unrecorded policy choices.

9.2 Govern map measure manage

The NIST AI Risk Management Framework organizes practice around Govern, Map, Measure and Manage. SRAI uses these functions as a disciplined risk loop: establish authority, understand context, evaluate performance and harm, then act on the results.

For generative AI, institutions also consider confabulation, information integrity, privacy, intellectual property, security and human over-reliance. Controls must match context and consequence.

9.3 Contestability and recourse

People affected by material decisions need understandable notice, routes to question outcomes and meaningful human review. Documentation must be usable by operational staff, auditors and decision makers, not only model developers.

Incident response includes detection, containment, communication, correction and learning. Serious incidents may trigger suspension or retirement rather than tuning alone.

Role	Accountability
Executive sponsor	Strategic authorization and resource commitment
Benefit owner	Outcome baseline, adoption and realized value
Domain owner	Professional validity and operational integration
Data owner and steward	Purpose, access, definitions and quality
Technical owner	Architecture, testing, operation and monitoring
Risk and assurance	Independent challenge, compliance and escalation

Executive implication

Record the decision, evidence, owner, uncertainty and next review date. Strategy becomes operational only when these elements are explicit.

10 From pilots to transformation

10.1 Evidence gates

A controlled lifecycle uses gates for problem validation, data qualification, risk acceptance, pilot evaluation and scale authorization. Each gate has an owner, evidence package and explicit decision.

Gates prevent activity from being mistaken for progress. A completed prototype is not evidence that users adopted it, that outcomes improved or that the institution can sustain it.

10.2 Pilot design

A pilot specifies comparator, population, metrics, duration, monitoring, human oversight and exit conditions. It tests operational integration as well as technical performance.

Pilot participants should understand the experimental status. Production data and consequential decisions require safeguards proportional to risk.

10.3 Scaling and retirement

Scaling requires service ownership, support, monitoring, budget, skills, security and change management. Performance drift, policy change or loss of value can justify redesign or retirement.

Retirement planning preserves records, handles data and interfaces safely, communicates changes and transfers users to an approved alternative.



Figure 3. Evidence gates convert experimentation into controlled institutional decisions.

Executive implication

Record the decision, evidence, owner, uncertainty and next review date. Strategy becomes operational only when these elements are explicit.

11 People skills and institutional change

11.1 Leadership literacy

Executives need enough AI literacy to question evidence, risk, cost and accountability. They do not need to become model developers. Their role is to create conditions for responsible decisions and to resist unsupported certainty.

Leadership training should use institutional cases, decision records and dashboards rather than generic demonstrations.

11.2 Counterpart capability

Capacity development includes domain experts, statisticians, data stewards, engineers, procurement officers, legal teams, auditors and operational managers. Training should be tied to responsibilities and real products.

Documentation, mentoring and supervised practice support transfer. Success is demonstrated when local teams can operate, diagnose and improve the system.

11.3 Adoption and incentives

People adopt systems that fit work, solve recognized problems and preserve legitimate professional judgement. Consultation should influence design, not merely communicate a completed decision.

Measures of adoption include appropriate use, exception handling, user confidence and realized outcomes, not login counts alone.

Executive implication

Record the decision, evidence, owner, uncertainty and next review date. Strategy becomes operational only when these elements are explicit.

12 Results value and learning

12.1 Results chains

A results chain links inputs and activities to outputs, outcomes and longer-term impact. AI performance belongs within this chain; accuracy is not the final institutional outcome.

Assumptions and external factors are recorded so that leaders can interpret results rather than attribute every change to the system.

12.2 Baselines indicators and equity

Indicators need definitions, sources, frequency, disaggregation, owners and targets. Distributional analysis checks whether average gains conceal exclusion or harm.

Where causal attribution is not feasible, the institution should state the strength of evidence honestly and combine quantitative measures with operational review.

12.3 Benefits reviews

Benefits reviews compare realized value with the approved hypothesis, including recurrent cost and unintended effects. They can release further investment, impose conditions or stop the initiative.

The executive dashboard should show outcomes, risk, adoption, cost, evidence confidence and pending decisions together.

Measure family	Example
Outcome	Reduction in release delay with quality maintained
Adoption	Appropriate use and exception handling
Risk	Material incidents and unresolved controls
Equity	Performance and access by relevant subgroup
Cost	Full recurrent cost per supported decision
Evidence confidence	Strength, recency and coverage of validation

Executive implication

Record the decision, evidence, owner, uncertainty and next review date. Strategy becomes operational only when these elements are explicit.

13 Experience-derived case portfolio

13.1 Case A Agricultural census modernization

A national statistics institution is preparing an agricultural census under financial, logistical and skills constraints. Field mapping, CAPI collection, quality control, tabulation and dissemination form one evidence chain. Potential AI uses include document retrieval for enumerators, anomaly triage, assisted coding and workload forecasting. None should bypass approved methodology or professional review.

The strategy begins by stabilizing classifications, frames, geospatial identifiers, synchronization controls and validation rules. A low-risk retrieval assistant may enter a controlled pilot while automated correction of official records remains deferred. Counterpart training and reproducible STATA or Python workflows are treated as core capability investments.

Decision recommendation. Executive decision: authorize discovery for bounded assistance, fund the data-quality foundation, and require methodological owners to approve any effect on official outputs.

Diagnostic question	Required evidence
What outcome is constrained?	Baseline, affected users and service standard
What must improve first?	Dependency and bottleneck assessment
Where can AI add value?	AI and non-AI comparator
Who is accountable?	Named outcome, data, technical and risk owners
What authorizes scale?	Passed gates and recorded executive approval

13.2 Case B Agricultural information warehouse

A ministry integrates primary surveys and administrative sources into a central warehouse accessible through a managed portal. Users at headquarters and in regions need standard and ad hoc reports. The historical transformation challenge is not simply database construction; it is agreement on definitions, ownership, refresh cycles and access.

AI readiness builds on this foundation. Semantic search, anomaly detection and forecasting are plausible only after metadata, lineage and stewardship are dependable. The case illustrates why data architecture is a strategic capability rather than an IT back-office concern.

Decision recommendation. Executive decision: prioritize governance and interoperability, then pilot services whose evidence can be traced to governed sources.

Diagnostic question	Required evidence
What outcome is constrained?	Baseline, affected users and service standard
What must improve first?	Dependency and bottleneck assessment
Where can AI add value?	AI and non-AI comparator
Who is accountable?	Named outcome, data, technical and risk owners
What authorizes scale?	Passed gates and recorded executive approval

13.3 Case C Results based programme monitoring

A cross-government programme has indicators in separate spreadsheets and systems. Reporting is late, baselines vary and responsibilities are unclear. AI-generated summaries appear attractive, but they would reproduce the fragmented evidence base.

The transformation sequence harmonizes the results framework, assigns indicator owners, establishes a warehouse and validates dashboards. AI is then tested for narrative synthesis and exception prioritization with links back to source evidence. Human programme managers approve interpretations and actions.

Decision recommendation. Executive decision: do not automate ambiguity; repair the results architecture and use AI only where traceability and review are demonstrable.

Diagnostic question	Required evidence
What outcome is constrained?	Baseline, affected users and service standard
What must improve first?	Dependency and bottleneck assessment
Where can AI add value?	AI and non-AI comparator
Who is accountable?	Named outcome, data, technical and risk owners
What authorizes scale?	Passed gates and recorded executive approval

13.4 Case D Food loss measurement and modelling

Multi-country food-loss research combines survey observations with environmental and socioeconomic variables. Predictive models may support targeting and scenario analysis, but sampling error, measurement differences and domain shift affect generalization.

The strategy requires explicit populations, comparable definitions, validation by country and uncertainty communication. A model that performs well overall may fail for a crop, geography or production system. Statistical design and domain expertise remain essential.

Decision recommendation. Executive decision: authorize modelling as decision support, retain survey evidence as the reference, and require subgroup validation before operational recommendations.

Diagnostic question	Required evidence
What outcome is constrained?	Baseline, affected users and service standard
What must improve first?	Dependency and bottleneck assessment
Where can AI add value?	AI and non-AI comparator
Who is accountable?	Named outcome, data, technical and risk owners
What authorizes scale?	Passed gates and recorded executive approval

14 Executive strategy clinic

The Ministry of Evidence and Rural Services produces agricultural statistics, administers support programmes and advises cabinet. Data arrive through surveys, administrative systems and regional spreadsheets. Leaders want an AI programme within twelve months. Quality reviews reveal inconsistent identifiers, uneven metadata, delayed validation and limited model-governance capacity. A vendor proposes a general platform with forecasting, document search and automated eligibility decisions.

Assignment

- Define three mission outcomes and their baselines.
- Construct a six-pillar readiness profile with evidence confidence.
- Classify six candidate initiatives as foundation, quick evidence, strategic bet, defer or prohibit.
- Define governance roles and decision gates.
- Prepare a 24-month roadmap with stop conditions.
- Write a two-page executive recommendation that distinguishes facts, assumptions and decisions.

Phase	Months	Required evidence	Decision
Mobilize	0-3	Mandate, owners, baseline, inventory	Approve discovery
Qualify	3-6	Data tests, risk map, comparators	Select pilots
Pilot	6-12	Operational evaluation and user evidence	Scale, redesign or stop
Institutionalize	12-18	Support, budget, controls and training	Authorize bounded scale
Review	18-24	Benefits, incidents and sustainability	Continue, expand or retire

15 Assessment

Item	Task	Marks
1	Explain why an AI strategy must begin with institutional outcomes rather than technologies.	10
2	Design an evidence rule for scoring data readiness.	10
3	Explain how dependencies can change initiative priority.	10
4	Allocate accountability for benefit, data, technical operation and residual risk.	10
5	Specify a comparator and stop condition for an anomaly-triage pilot.	10
6	Define an outcome indicator and an equity disaggregation.	10
7	Recommend which initiative in the clinic should be piloted first and defend the choice.	20
8	Draft a decision paragraph that separates evidence, uncertainty and authorization.	20

Instructor solutions and marking guide

1 Strategy formulation

Explain why an AI strategy must begin with institutional outcomes rather than technologies.

Model answer. A strong answer connects mandate, beneficiary, baseline and owned outcome; explains that technology is a mechanism; and identifies opportunity cost and accountability.

Marking: 40 percent for correct concepts, 30 percent for evidence and reasoning, 20 percent for institutional applicability, and 10 percent for clarity.

2 Readiness

Design an evidence rule for scoring data readiness.

Model answer. Require named sources, quality tests, population coverage, lineage, recency and confidence. Missing critical evidence should lower confidence or block the score rather than be imputed silently.

Marking: 40 percent for correct concepts, 30 percent for evidence and reasoning, 20 percent for institutional applicability, and 10 percent for clarity.

3 Portfolio

Explain how dependencies can change initiative priority.

Model answer. Dependencies reveal enabling investments and bottlenecks. A lower-visibility data or governance foundation may precede a high-value service because the latter cannot be validated or operated without it.

Marking: 40 percent for correct concepts, 30 percent for evidence and reasoning, 20 percent for institutional applicability, and 10 percent for clarity.

4 Governance

Allocate accountability for benefit, data, technical operation and residual risk.

Model answer. The benefit owner owns the outcome; the data owner and steward govern purpose and quality; the technical owner operates the system; the authorized executive or risk owner accepts residual exposure with independent assurance.

Marking: 40 percent for correct concepts, 30 percent for evidence and reasoning, 20 percent for institutional applicability, and 10 percent for clarity.

5 Pilot

Specify a comparator and stop condition for an anomaly-triage pilot.

Model answer. Compare triage with the current quality-review process using recall, precision, workload, missed critical errors and subgroup performance. Stop if critical misses exceed the approved threshold or reviewers cannot trace recommendations.

Marking: 40 percent for correct concepts, 30 percent for evidence and reasoning, 20 percent for institutional applicability, and 10 percent for clarity.

6 Value

Define an outcome indicator and an equity disaggregation.

Model answer. Define baseline, numerator, denominator, source, frequency, owner and target. Disaggregate by geography, service channel or relevant beneficiary group and test whether average improvement conceals deterioration.

Marking: 40 percent for correct concepts, 30 percent for evidence and reasoning, 20 percent for institutional applicability, and 10 percent for clarity.

7 Case analysis

Recommend which initiative in the clinic should be piloted first and defend the choice.

Model answer. A defensible choice is bounded document retrieval or quality triage after data qualification, because it can produce evidence without delegating consequential authority. The answer must state dependencies, owner and gate.

Marking: 40 percent for correct concepts, 30 percent for evidence and reasoning, 20 percent for institutional applicability, and 10 percent for clarity.

8 Executive writing

Draft a decision paragraph that separates evidence, uncertainty and authorization.

Model answer. The paragraph should state verified facts, unresolved assumptions, material risks, the authorized scope, conditions, accountable owner and next review date.

Marking: 40 percent for correct concepts, 30 percent for evidence and reasoning, 20 percent for institutional applicability, and 10 percent for clarity.

Appendix A Applied field guide

These field notes convert recurring lessons from international statistics, information-system, monitoring and evaluation, and capacity-development assignments into practical AI transformation controls. Each note is designed as a one-page discussion and review aid.

A1 Entering an institution through its mandate

An inception mission should begin with the legal mandate, strategic plan, current programme and decisions that leadership must make. In census and statistical assignments, requests often arrive as technical tasks - sampling, CAPI, mapping, tabulation or dashboards - while the underlying need is institutional: produce credible evidence on time with available resources and accountable national ownership. The consultant must connect the requested output to that wider obligation.

A practical inception record states the decision problem, expected users, authoritative sources, constraints, dependencies and approval route. It also identifies claims that remain unverified. This prevents early enthusiasm from becoming an implicit commitment to a platform or model.

Executive field checklist

- State the institutional decision and accountable owner.
- Identify the evidence already verified and the evidence still missing.
- Specify the capability, safeguard or dependency that must be in place.
- Record the authorization, condition, next review date and stopping rule.

Reflection question

What would change in the recommended decision if the weakest evidence proved incorrect or the responsible team could not sustain the proposed control?

A2 Validating the diagnosis with counterparts

Diagnostics gain authority when national counterparts recognize the process map, evidence gaps and constraints as accurate. Validation workshops should therefore test the representation of reality, not seek ceremonial endorsement. Participants should be able to correct ownership, identify informal workarounds and explain why documented procedures differ from practice.

The facilitator records agreements, unresolved issues and the evidence needed to close them. This method, used in strategic-planning and information-system assignments, reduces resistance because the baseline becomes a jointly governed product rather than an external judgement.

Executive field checklist

- State the institutional decision and accountable owner.
- Identify the evidence already verified and the evidence still missing.
- Specify the capability, safeguard or dependency that must be in place.
- Record the authorization, condition, next review date and stopping rule.

Reflection question

What would change in the recommended decision if the weakest evidence proved incorrect or the responsible team could not sustain the proposed control?

A3 Converting strategic plans into decision architecture

Many strategic plans contain ambitions, activities and indicators but do not specify decision rights or evidence gates. An AI transformation plan must add those operational elements. Each objective needs a benefit owner, capability dependencies, financing assumptions, review dates and thresholds for continuation.

The transformation office should maintain one portfolio register that connects objectives, initiatives, risks, budgets and results. This is more useful than separate innovation lists, procurement plans and monitoring tables that cannot be reconciled.

Executive field checklist

- State the institutional decision and accountable owner.
- Identify the evidence already verified and the evidence still missing.
- Specify the capability, safeguard or dependency that must be in place.
- Record the authorization, condition, next review date and stopping rule.

Reflection question

What would change in the recommended decision if the weakest evidence proved incorrect or the responsible team could not sustain the proposed control?

A4 Protecting statistical quality in AI assisted workflows

Official statistics depend on concepts, classifications, frames, sampling designs, edit rules, estimation methods and release procedures. AI may assist coding, triage, retrieval or forecasting, but it must not obscure methodological responsibility. A recommendation should be traceable to the input, model version and professional review.

Quality assurance should examine error by relevant domain, not only overall accuracy. In agricultural work this may include crop, production system, region, season or holding type. A model that performs well nationally can still be unsuitable for a small but policy-critical population.

Executive field checklist

- State the institutional decision and accountable owner.
- Identify the evidence already verified and the evidence still missing.
- Specify the capability, safeguard or dependency that must be in place.
- Record the authorization, condition, next review date and stopping rule.

Reflection question

What would change in the recommended decision if the weakest evidence proved incorrect or the responsible team could not sustain the proposed control?

A5 Designing for field realities

Field systems operate under connectivity, device, language, travel, supervision and time constraints. Strategy written only from headquarters can underestimate the operational burden of new controls or tools. Field testing should observe how enumerators and supervisors actually use instructions, resolve exceptions and synchronize records.

An AI assistant for field staff should use approved material, work within known connectivity limits, expose uncertainty and route unresolved questions to a human supervisor. Its value is reduced search time and more consistent support, not autonomous reinterpretation of methodology.

Executive field checklist

- State the institutional decision and accountable owner.
- Identify the evidence already verified and the evidence still missing.
- Specify the capability, safeguard or dependency that must be in place.
- Record the authorization, condition, next review date and stopping rule.

Reflection question

What would change in the recommended decision if the weakest evidence proved incorrect or the responsible team could not sustain the proposed control?

A6 Building the governed data chain

Data warehouses and analytical platforms create a governed chain from source to decision product. Experience with primary surveys, administrative sources and indicator databases shows that integration requires shared identifiers, classifications, refresh rules, validation and stewardship. Infrastructure alone does not create comparability.

The AI strategy should identify which datasets are authoritative, which transformations are reproducible and which uses are permitted. Evidence lineage must survive export to dashboards, models and narrative products so that users can trace and challenge results.

Executive field checklist

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A7 Designing dashboards for decisions

A dashboard should correspond to a decision rhythm. Operational managers need exceptions and workloads; programme managers need progress and bottlenecks; executives need outcomes, risks, resources and pending authorizations. One crowded dashboard rarely serves all three levels.

Every measure needs a definition, baseline, owner, update cycle and action rule. AI-generated commentary can help summarize changes, but it must link back to validated indicators and distinguish observation from interpretation.

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A8 Moving from pilot evidence to scale

Pilots frequently demonstrate technical possibility without testing institutional operation. A credible pilot includes users, workflow, support, data refresh, risk controls, comparison and outcome measures. It also consumes realistic staff time and records exceptions.

Before scale, leadership should see evidence on adoption, failure modes, recurrent cost, security, subgroup performance and support capacity. A positive demonstration is only one input. The scale decision is a new authorization with a broader risk and resource envelope.

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Reflection question

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A9 Procuring capability rather than dependency

Procurement documents should describe outcomes, interfaces, evidence obligations, security, data rights, documentation, knowledge transfer and exit arrangements. Requirements written around a preferred product weaken strategic choice and may create lock-in.

Vendors should demonstrate how the institution will test performance, obtain records, manage updates and migrate or retire the service. Contract acceptance should depend on validated evidence products and counterpart capability, not installation alone.

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A10 Making training part of the operating model

Training is effective when it maps to named roles and real tasks. Executives need decision literacy; domain teams need validation and exception skills; technical teams need operation and monitoring; stewards need quality and access procedures; auditors need evidence access. A single generic AI course cannot meet these needs.

Capacity plans should include supervised practice, documentation, mentoring and an assessment of independent performance. The objective is not attendance but the ability of local teams to operate, question and improve the system.

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Reflection question

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A11 Managing international and subnational variation

Multi-country and decentralized programmes must expect variation in definitions, infrastructure, institutions and production systems. Standardization is valuable where concepts and interfaces must align, but local adaptation remains necessary for language, workflow and risk.

The strategy should define a controlled core and documented extensions. Comparative analysis should not treat missing or non-comparable data as if they were equivalent observations. Local validation is essential before transferring models or thresholds.

Executive field checklist

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Reflection question

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A12 Communicating uncertainty

Executives need uncertainty translated into decision consequences. Confidence intervals, scenario ranges and sensitivity results should answer whether the decision is robust, what evidence would reduce uncertainty and what exposure remains if action proceeds now.

A dashboard that reports one score without assumptions invites false certainty. Decision papers should show contested rankings, missing evidence and the conditions under which the recommendation would change.

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Reflection question

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A13 Aligning ambition with financing

AI programmes incur costs beyond development: data preparation, integration, security, monitoring, licences, computing, user support, evaluation and renewal. A pilot financed by a temporary project can create an unfunded operational obligation.

The roadmap should identify full recurrent cost, funding owner and procurement lead time. High-value opportunities may be sequenced behind foundational investments when financing or capability cannot sustain them safely.

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Reflection question

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A14 Continuous institutional learning

Transformation is not complete at deployment. Institutions need scheduled benefits reviews, incident analysis, model and data monitoring, user feedback and authority to revise controls. Learning must be connected to budget and portfolio decisions.

A mature programme preserves evidence from stopped initiatives as well as successful ones. This institutional memory improves future selection and counters the tendency to repeat attractive but unsupported pilots.

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A15 Closing a consulting engagement responsibly

A responsible closeout transfers documents, code, credentials, data definitions, operating procedures, decision registers and unresolved risks to named owners. It verifies that counterparts can perform critical tasks and identifies support that remains necessary.

The final report should distinguish delivered capability from planned capability. It should record limitations plainly and provide prioritized next decisions. This protects the institution from assuming that a project milestone equals sustainable transformation.

Executive field checklist

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16 Conclusion

AI strategy is the disciplined allocation of institutional attention, authority and resources to evidence-backed opportunities. Its quality is visible in what leadership declines, defers and stops as much as in what it funds. Mission alignment, qualified data, responsible governance, capable people and measurable outcomes form one system. The transformation leader's obligation is to preserve accountable human authority while creating room for controlled learning.

Ten executive principles

- Start with mandate and beneficiary value.
- Diagnose before prescribing.
- Compare AI with simpler interventions.
- Treat data quality and lineage as strategic controls.
- Keep risk gates separate from weighted scores.
- Name outcome, data, technical and risk owners.
- Pilot the operating model, not only the model.
- Build counterpart capability from the beginning.
- Measure adoption, outcomes, equity, cost and incidents together.
- Scale, redesign or retire through recorded executive decisions.

Glossary

Term	Meaning
AI initiative	A governed intervention using AI to improve a defined outcome.
Benefit owner	Leader accountable for realizing an approved outcome.
Decision gate	Formal evidence review that authorizes progression, redesign or stop.
Evidence confidence	Assessment of the strength, recency and coverage of supporting evidence.
Readiness	Use-case-specific ability to develop, govern, operate and sustain an intervention.
Residual risk	Risk remaining after controls and requiring explicit acceptance.
Transformation thesis	Causal explanation linking institutional change to expected value.

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